What we learned from the two studies

Card study

Fraud clusters in short periods, and the amount 1.00 appears repeatedly. But that amount also appears in 13,575 legitimate payments. Without card IDs, we cannot tell whether the small payments test a card before a larger purchase.

Financial transactions

Hour, city and payment category alone do little to distinguish fraud. Small-then-large sequences cover only 168 of 179,553 fraud cases. Flagging city changes would select 71.8% of all payments to cover 75.1% of fraud cases: a lot of review for little extra targeting.

What to do next

Link payments to reliable account or card histories and include failed attempts. Test combined signals on later transactions. Judge them by extra fraud found and legitimate customers affected. Payment value helps set review priorities; it does not by itself prove fraud.

We cannot say whether behaviour changed by 2026 or stayed the same. The card file has no verified transaction year, and the two populations differ.

A large payment is a stronger clue in one file than in the other

We compared the largest 10% of payments with the rest of each file. This avoids treating the same cash amount as equivalent across two different populations.

Group	Fraud / payments	Fraud rate
Card study: largest 10%	**85 / 28,483**	**0.298%**
Card study: remaining payments	407 / 256,324	0.159%
Financial transactions: largest 10%	**17,999 / 500,004**	**3.600%**
Financial transactions: remaining	161,554 / 4,499,996	3.590%

In the card study, the largest payments have 1.88 times the fraud rate of the rest. In the financial-transactions study, size makes almost no difference.

That tells us how useful amount is in each file. It does not show that offenders changed their behaviour by 2026.

A repeated amount is a lead. It is an expensive rule on its own.

120 legitimate payments for every fraud case

That is what a review of every 1.00 payment would produce in the card study: 13,575 legitimate payments and 113 fraud cases. It would still miss 379 of the 492 fraud cases.

City change

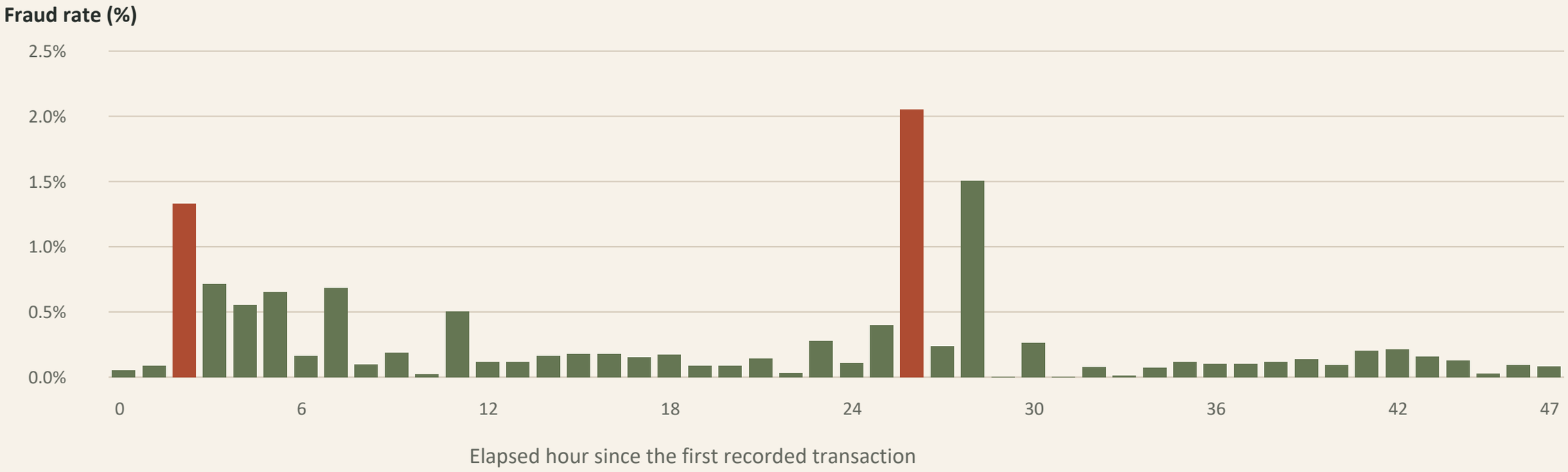
In the financial-transactions file, this would flag 71.8% of all payments to cover 75.1% of fraud cases. The queue would include 3,454,371 legitimate payments.

Small then large

This sequence appears 4,126 times, with 168 fraud labels. It covers fewer than 1 in 1,000 fraud cases in that file. It cannot be the main detection rule.

There are bursts in the card study. We cannot give them a verified clock time.

The source gives elapsed seconds. Calling both highlighted periods 02:00 depends on the assumed midnight start.



Hour 2: 21 frauds in 1,576 payments (1.33%). Hour 26: 36 in 1,752 (2.05%). These are useful periods to investigate within this two-day file.

The financial-transactions file has a much flatter hourly profile: 3.51%-3.66% fraud. Different time origins and observation windows prevent a reliable comparison of clock-time behaviour.

What this should change in the next review

01 **Link the activity before calling it a pattern.**

For repeated small payments, obtain card tokens, merchant/session links and failed attempts. Test limits or extra checks on suspicious bursts, then count the legitimate customers affected.

02 **Use city and amount as context.**

A city change alone selects too much of the financial-transactions file. Check it against the sender's own history. For online card purchases, an extra 3DS check is one option to test.

03 **Measure what a new check actually adds.**

Use later transactions. Compare extra fraud found, legitimate payments challenged and net loss after recoveries. The current files do not measure how much fraud a control would prevent.

Did behaviour change by 2026? These files cannot answer that yet. We need verified 2026 transactions from a comparable population.

What the figures mean

The two source files

Card study: creditcard.csv, 284,807 payments and 492 fraud labels over 47h 59m 52s. No verified calendar year or card/account IDs. Identical rows remain in the analysis.
Financial-transactions study: financial_fraud_detection_dataset.csv, 5,000,000 payments and 179,553 fraud labels, dated 1 January 2023 to 1 January 2024. Only one day of 2024 is present.

Calculations used here

Fraud rate = fraud labels / payments in the stated group. Largest 10% starts at 203.00 in the card file and 1,126.50 in the financial file. Ties stay included: 28,483 and 500,004 payments. Each group is compared with the rest of its own file. Currency and customer populations are not confirmed as equivalent.

Small then large = previous sender payment below 100, followed by one above 1,000 within 24 hours. Sender histories are sorted by timestamp. A first observation has no previous city. All rule examples count current payments, not prevented fraud or individual offenders.

Before using the financial file to build rules

Supplied time intervals include negatives and label-linked missingness. Device hashes change on every consecutive observed sender pair. All fraud-type labels say card_not_present. Confirm how these fields and labels were created.

Control guidance: EMVCo - EMV 3-D Secure

Stripe - card-testing controls